2021 AIME II Problems/Problem 6

Contents

Problem

For any finite set S, let $|S|$ denote the number of elements in S. Find the number of ordered pairs (A, B) such that A and B are (not necessarily distinct) subsets of $\{1, 2, 3, 4, 5\}$ that satisfy

$$|A| \cdot |B| = |A \cap B| \cdot |A \cup B|$$

Solution 1

By PIE, $|A| + |B| - |A \cap B| = |A \cup B|$. Substituting into the equation and factoring, we get that $(|A| - |A \cap B|)(|B| - |A \cap B|) = 0$, so therefore $A \subseteq B$ or $B \subseteq A$. WLOG $A \subseteq B$, then for each element there are 3 possibilities, either it is in both A and B, it is in B but not A, or it is in neither A nor B. This gives us 3^5 possibilities, and we multiply by 2 since it could have also been the other way around. Now we need to subtract the overlaps where $A = B$, and this case has $2^5 = 32$ ways that could happen. It is 32 because each number could be in the subset or it could not be in the subset. So the final answer is $2 \cdot 3^5 - 2^5 = \boxed{454}$.

~math31415926535

Solution 2

We denote $\Omega = \{1, 2, 3, 4, 5\}$. We denote $X = A \cap B$, $Y = A \backslash (A \cap B)$, $Z = B \backslash (A \cap B)$, $W = \Omega \backslash (A \cup B)$.

Therefore, $X \cup Y \cup Z \cup W = \Omega$ and the intersection of any two out of sets X, Y, Z, W is an empty set. Therefore, (X, Y, Z, W) is a partition of Ω.

Following from our definition of X, Y, Z, we have $A \cup B = X \cup Y \cup Z$.

Therefore, the equation

$$|A| \cdot |B| = |A \cap B| \cdot |A \cup B|$$

can be equivalently written as

$$\left(|X|+|Y|\right)\left(|X|+|Z|\right)=|X|\left(|X|+|Y|+|Z|\right).$$

This equality can be simplified as

$$|Y|\cdot|Z|=0.$$

Therefore, we have the following three cases: (1) $|Y|=0$ and $|Z|\neq 0$, (2) $|Z|=0$ and $|Y|\neq 0$, (3) $|Y|=|Z|=0$. Next, we analyze each of these cases, separately.

Case 1: $|Y|=0$ and $|Z|\neq 0$.

In this case, to count the number of solutions, we do the complementary counting.

First, we count the number of solutions that satisfy $|Y|=0$.

Hence, each number in Ω falls into exactly one out of these three sets: X, Z, W. Following from the rule of product, the number of solutions is 3^5.

Second, we count the number of solutions that satisfy $|Y|=0$ and $|Z|=0$.

Hence, each number in Ω falls into exactly one out of these two sets: X, W. Following from the rule of product, the number of solutions is 2^5.

Therefore, following from the complementary counting, the number of solutions in this case is equal to the number of solutions that satisfy $|Y|=0$ minus the number of solutions that satisfy $|Y|=0$ and $|Z|=0$, i.e., 3^5-2^5.

Case 2: $|Z|=0$ and $|Y|\neq 0$.

This case is symmetric to Case 1. Therefore, the number of solutions in this case is the same as the number of solutions in Case 1, i.e., 3^5-2^5.

Case 3: $|Y|=0$ and $|Z|=0$.

Recall that this is one part of our analysis in Case 1. Hence, the number solutions in this case is 2^5.

By putting all cases together, following from the rule of sum, the total number of solutions is equal to

$$\begin{aligned}\left(3^5-2^5\right)+\left(3^5-2^5\right)+2^5 &= 2\cdot 3^5-2^5\\ &=\boxed{454}.\end{aligned}$$

~ Steven Chen (www.professorchenedu.com)

Solution 3 (Principle of Inclusion-Exclusion)

By the **Principle of Inclusion-Exclusion (abbreviated as PIE)**, we have $|A\cup B|=|A|+|B|-|A\cap B|,$ from which we rewrite the given equation as

$$|A|\cdot|B|=|A\cap B|\cdot\left(|A|+|B|-|A\cap B|\right).$$

Rearranging and applying Simon's Favorite Factoring Trick give

$$\begin{aligned}|A|\cdot|B| &= |A\cap B|\cdot|A|+|A\cap B|\cdot|B|-|A\cap B|^2\\ |A|\cdot|B|-|A\cap B|\cdot|A|-|A\cap B|\cdot|B| &= -|A\cap B|^2\\ \left(|A|-|A\cap B|\right)\cdot\left(|B|-|A\cap B|\right) &= 0,\end{aligned}$$

from which at least one of the following is true:

- $|A| = |A \cap B|$
- $|B| = |A \cap B|$

Let $|A \cap B| = k.$ For each value of $k \in \{0,1,2,3,4,5\},$ we will use PIE to count the ordered pairs $(A,B):$

Suppose $|A| = k.$ There are $\binom{5}{k}$ ways to choose the elements for $A.$ These k elements must also appear in $B.$ Next, there are 2^{5-k} ways to add any number of the remaining $5-k$ elements to B (Each element has 2 options: in B or not in $B.$). There are $\binom{5}{k}2^{5-k}$ ordered pairs for $|A| = k.$ Similarly, there are $\binom{5}{k}2^{5-k}$ ordered pairs for $|B| = k.$

To fix the overcount, we subtract the number of ordered pairs that are counted twice, in which $|A| = |B| = k.$ There are $\binom{5}{k}$ such ordered pairs.

Therefore, there are

$$2\binom{5}{k}2^{5-k} - \binom{5}{k}$$

ordered pairs for $|A \cap B| = k.$

Two solutions follow from here:

Solution 3.1 (Binomial Theorem)

The answer is

$$\begin{aligned}\sum_{k=0}^{5}\left[2\binom{5}{k}2^{5-k} - \binom{5}{k}\right] &= 2\sum_{k=0}^{5}\binom{5}{k}2^{5-k} - \sum_{k=0}^{5}\binom{5}{k} \\ &= 2(2+1)^5 - (1+1)^5 \\ &= 2(243) - 32 \\ &= \boxed{454}.\end{aligned}$$

~MRENTHUSIASM

Solution 3.2 (Bash)

The answer is

$$\begin{aligned}&\sum_{k=0}^{5}\left[2\binom{5}{k}2^{5-k} - \binom{5}{k}\right] \\ &= \left[2\binom{5}{0}2^{5-0} - \binom{5}{0}\right] + \left[2\binom{5}{1}2^{5-1} - \binom{5}{1}\right] + \left[2\binom{5}{2}2^{5-2} - \binom{5}{2}\right] + \left[2\binom{5}{3}2^{5-3} - \binom{5}{3}\right] + \left[2\binom{5}{4}2^{5-4} - \binom{5}{4}\right] + \left[2\binom{5}{5}2^{5-5} - \binom{5}{5}\right] \\ &= \left[2\,(1)\,2^5 - 1\right] + \left[2\,(5)\,2^4 - 5\right] + \left[2\,(10)\,2^3 - 10\right] + \left[2\,(10)\,2^2 - 10\right] + \left[2\,(5)\,2^1 - 5\right] + \left[2\,(1)\,2^0 - 1\right] \\ &= 63 + 155 + 150 + 70 + 15 + 1 \\ &= \boxed{454}.\end{aligned}$$

~MRENTHUSIASM

Solution 4 (Simple Bash)

Proceed with Solution 1 to get $(|A|-|A\cap B|)(|B|-|A\cap B|)=0$. WLOG, assume $|A|=|A\cap B|$. Thus, $A\subseteq B$.

Since $A\subseteq B$, if $|B|=n$, there are 2^n possible sets A, and there are also $\binom{5}{n}$ ways of choosing such B.

Therefore, the number of possible pairs of sets (A,B) is

$$\sum_{k=0}^{5} 2^n \binom{5}{n}$$

We can compute this manually since it's only from $k=0$ to 5, and computing gives us 243. We can double this result for $B\subseteq A$, and we get $2(243)=486$.

However, we have double counted the cases where A and B are the same sets. There are 32 possible such cases, so we subtract 32 from 486 to get $\boxed{454}$.

~ adam_zheng

Video Solution by Interstigation

https://youtu.be/jEghPVjyHoc

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Video Solution & Set Theory Review

https://youtu.be/3vcLujj74RM

~MathProblemSolvingSkills.com

See Also

2021 AIME II (**Problems** • **Answer Key** • Resources)

Preceded by **Problem 5** | Followed by **Problem 7**

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